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The EUV Snapshot Imaging Spectrograph

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1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as: $I(x, y, \lambda, t)$, where x and y are the helioprojective Cartesian coordinates (W. T. Thompson 2006), λ is wavelength, and t is time. The ideal solar imaging spectrograph would capture $I(x, y, \lambda, t)$ with high resolution in x, y, λ, and t *and* over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time t_0: $I(x, y, \lambda, t_0)$. Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than ∼150 nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker lines within them bound the velocity resolution near $\sim 1000\,\mathrm{km\,s^{-1}}$ (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records λ against y without ambiguity, and x is recovered by stepping the slit across the target. Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity: neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along x. Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the recently flown Solar eruptioN Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024) demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once, so every emission line forms an image of the Sun and those images superimpose on the detector. The result-

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ing frame, an *overlappogram*, encodes x and λ along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as space-borne solar spectroscopy, having been recorded by the Naval Research Laboratory's (NRL's) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed $m = 0$ order and the $m = \pm1$ orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a plane, they fill the cylindrical volume of a sounding rocket payload poorly.

might just need to go through the whole paper then write this paragraph The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary mask, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section **??** states the scientific objectives and the requirements they impose. Section **??** describes the instrument itself, and Section **??** the mission profile.

2. THE ESIS CONCEPT

ESIS and MOSES answer the same question in the same way. Both disperse the whole field of view without a slit, both record several images of it simultaneously, and both depend on an inversion to recover a line profile at every point of that field. Where they differ is in how the projections are arranged, and that difference is the accumulated result of two MOSES flights. This section sets out what those flights revealed about the limits of the MOSES design, and how each limit shaped ESIS.

2.1. *Limitations of the MOSES Design*

A single concave grating serves as the imaging element of MOSES, forming three images on three charge-coupled devices (CCDs) (J. L. Fox et al. 2010), and a flat secondary folds the optical path in half. One consequence of that arrangement is convenient: with the cameras correctly placed, the undispersed order can be focused in visible light, so the whole telescope is aligned without a source at the operating wavelength. The rest of the consequences are constraints.

Because the fold mirror provides no magnification, the instrument cannot be made shorter than half the 5 m focal length of the grating (J. L. Fox et al. 2010; J. L. Fox 2011). Dispersion is likewise geometric: it is set by how far apart the cameras sit, so reaching the maximum $29\,\mathrm{km\,s^{-1}}$ per pixel (J. L. Fox et al. 2010) means pushing the outboard cameras to the edge of a payload only 0.56 m across. Both constraints act in a single plane, which leaves the cylindrical volume of the payload largely unused along the orthogonal directions.

That same plane limits how many images can be recorded at all. Three diffraction orders, $m = -1$, 0, and $+1$, are as many as the payload will hold, so MOSES measures three numbers at each point of the field. Three numbers will not constrain a line profile beyond three degrees of freedom, which in practice means intensity, Doppler shift, and width (C. C. Kankelborg & R. J. Thomas 2001), and only compact, well isolated features have yielded more (J. L. Fox et al. 2010; T. Rust & C. C. Kankelborg 2019). This is the familiar behavior of any tomographic measurement, where the detail recoverable in the object follows from the number of viewing angles available (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008): finer spectral structure, whether an additional line in the passband or a higher moment of the profile, requires more projections than MOSES can provide.

Having only one plane of dispersion also makes the instrument sensitive to the orientation of the scene. The transition region and corona are threaded by magnetic field, and the emission is drawn out into loops and filaments along it (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). A filament lying nearly perpendicular to the dispersion acts as its own spectrograph slit, and recovering a Doppler shift reduces to triangulation between the orders (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018), with sufficiently isolated features yielding even the splitting of a double-peaked profile (T. Rust & C. C. Kankelborg 2019). A filament lying nearly parallel to the dispersion yields none of this, and since the field has no globally preferred direction, which case applies is a matter of where the instrument happens to be pointed.

A single grating must also form all three images at once, which leaves too few degrees of freedom in the optical prescription to control aberrations in every order. MOSES flew its first mission slightly defocused (T. Rust & C. C. Kankelborg 2019), which widened the point spread functions to several pixels and made them differ between orders, the outboard ones elongated along different axes (T. Rust & C. C. Kankelborg 2019; S. Atwood & C. Kankelborg 2018). Structure introduced by the instrument in this way is difficult to separate from structure in the spectrum, and it complicates the inversion (S. Atwood & C. Kankelborg 2018; H. T. Courrier & C. C. Kankelborg 2018).

Without a field stop, the field of view is defined only by the geometry of the optics, and differs from order to order and with wavelength. Light from outside the intended field reaches the detectors, and it is not negligible: comparing synthetic images against the flight data, J. D. Parker & C. C. Kankelborg (2022) attributed roughly a tenth of the undispersed image to more than ten faint lines in the passband, most of them invisible in the dispersed orders. An undispersed channel is attractive precisely because it is free of spatial-spectral ambiguity, but that result shows the ambiguity is replaced by a spectral one.

Finally, the detectors must be read out before they can be exposed again, and the roughly 6 s this takes (J. L. Fox 2011) is expensive when a sounding rocket observes for about five minutes in total. Half of the observing time is spent reading rather than collecting, and the exposures that remain must be traded against one another, which is why MOSES flew a sequence ranging from a quarter of a second to 24 s.

Taken together, our experience with MOSES identifies eight limitations of the design:

1. inefficient use of the payload volume
2. dispersion constrained by the payload diameter
3. too few dispersed images
4. a single plane of dispersion
5. insufficient control of aberrations
6. a field of view which is neither sharply defined nor the same in every order
7. spectral contamination from outside the intended field
8. an observing duty cycle of only half the flight

Each of these shaped the design of ESIS.

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